# IVAN LARININ

**Game Developer • Unreal Engine & Unity Gameplay / Tools Programmer**

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## PROFESSIONAL SUMMARY

Versatile Game Developer and Tools Programmer specializing in high-performance gameplay systems, custom engine tooling, and procedural pipeline integration in Unreal Engine (C++) and Unity (C#). Proven track record developing robust Slate-based node editors, multi-root DAG runtime subsystems, compute shaders, and artist-facing editor extensions. Combines deep procedural engineering expertise (Houdini C++ SOPs / HDAs, VFX pipeline on high-profile film productions) with real-time systems architecture to streamline content creation and empower multidisciplinary teams.

## CORE COMPETENCIES

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| **Engines:** | Unreal Engine 5 (UE5 C++, Blueprint, Slate UI, Subsystems), Unity (C#, URP, Editor Scripting) |
| **Languages:** | C++, C#, HLSL, GLSL, Python, VEX, OpenCL |
| **Graphics & Tools:** | Compute Shaders, Vertex Animation Textures (VAT), GPU Instancing, Tessellation, Houdini Engine HDAs, Git, Perforce |
| **Core Concepts:** | Editor Tooling, Runtime Subsystems, Custom Graph Editors, Procedural World Generation, Memory & Cache Optimization |

## FEATURED GAME DEVELOPMENT & TOOL PROJECTS

**UE5 Quest & Dialogue Graph System Plugin** — *C++, Unreal Engine 5, Slate UI*

Engineered an extensible C++ plugin for Unreal Engine 5 featuring an integrated Slate visual node graph editor toolkit and a runtime quest manager subsystem (UWorldSubsystem). Implemented multi-root DAG execution logic, dynamic condition evaluation, and robust JSON/binary state serialization for clean quest tracking and mission scripting.

**Real-Time Ocean Simulation & Dynamic Shading** — *Compute Shaders, HLSL, C#, Unity*

Built a high-performance ocean simulation in Unity decoupling simulation cost from geometry by moving Gerstner wave calculations to a Compute Shader. Generated global displacement and normal render textures dynamically, integrated distance-based tessellation, and implemented camera-near-plane water slicing for underwater transitions.

**Terrain Prefab Scatter Brush Tool** — *C#, Unity Editor Tools*

Developed an intuitive custom Unity EditorWindow enabling artists to paint gameobject prefabs directly onto terrains and static geometry in the Scene View. Integrated real-time surface normal raycast alignment, customizable radius and density controls, keyboard shortcuts, and full Undo/Redo stack integration.

**Procedural Environment & City Generators** — *Houdini Engine, Unreal Engine 5, Unity*

Constructed interactive procedural toolsets using Houdini Engine for Unreal and Unity. Designed modular procedural city and architectural layout tools for rapid level blockout, along with rock arch and terrain dressing generators that expose parameters directly within the engine viewports.

**GPU-Instanced Foliage Tool & Caustics Rendering** — *Unity URP, HLSL, Houdini VAT*

Developed a GPU-instanced underwater vegetation system leveraging Houdini Vertex Animation Textures (VAT) for zero-CPU bone deformation overhead. Authored custom Universal Render Pipeline (URP) projector shaders for caustics with depth reconstruction, bounding box clipping, and chromatic aberration.

## PROFESSIONAL EXPERIENCE

**Game Tools & Technical Artist / Developer** *2021 – Present*

**Independent / Studio Contract | Montreal, QC**

Designs custom editor extensions, gameplay subsystems, and procedural content pipelines for Unreal Engine and Unity. Partners closely with level designers and 3D artists to eliminate production bottlenecks through bespoke C++ and C# tools, automation scripts, and custom real-time shaders.

**Senior Technical FX & Pipeline Specialist** *Feature Film & High-End Production*

**Major VFX Studios | Credits include: *Dune (Part One)***

Engineered custom C++ Houdini Surface Operators (SOPs), OpenCL solvers, and procedural simulation systems for large-scale environmental dynamics on major international productions. Built robust data exchange bridges and automated ingestion pipelines between procedural DCC packages and production rendering pipelines.